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Exploration of empirical Bayes hierarchical modeling for the analysis of genome-wide association study data

ELIZABETH A. HERON*, COLM O'DUSHLAINE, RICARDO SEGURADO,
LOUISE GALLAGHER, MICHAEL GILL

Neuropsychiatric Genetics Research Group and Department of Psychiatry, Trinity College Dublin, Trinity Centre for Health Sciences, James's Street, Dublin 8, Ireland

eaheron@tcd.ie

Abstract

In the analysis of genome-wide association (GWA) data, the aim is to detect statistical associations between single nucleotide polymorphisms (SNPs) and the disease or trait of interest. These SNPs, or the particular regions of the genome they implicate, are then considered for further study. We demonstrate through a comprehensive simulation study that the inclusion of additional, biologically relevant information through a 2-level empirical Bayes hierarchical model framework offers a more robust method of detecting associated SNPs. The empirical Bayes approach is an objective means of analyzing the data without the need for the setting of subjective parameter estimates. This framework gives more stable estimates of effects through a reduction of the variability in the usual effect estimates. We also demonstrate the consequences of including additional information that is not informative and examine power and false-positive rates. We apply the methodology to a number of genome-wide association (GWA) data sets with the inclusion of additional biological information. Our results agree with previous findings and in the case of one data set (Crohn's disease) suggest an additional region of interest. Coronary artery disease; Crohn's disease; Multilevel model; Rheumatoid arthritis; Semi-Bayes; Type 2 diabetes.

1 Introduction

Genome-wide association studies (GWASs) aiming to detect associations between observable traits and genetic variation across the genome are now commonplace thanks to the advent of several economically feasible high-throughput genotyping technologies. This genetic variation is typically in the form of single nucleotide polymorphisms (SNPs). Data consist of genotypes at each of the SNPs, for members of groups of cases and controls or family members (parents and affected offspring). The choice of case-control or family data may depend on the age-of-onset of the disease/trait being studied. There are many approaches to analyzing these data, including, for example, single SNP analyses and pathway approaches; allison2006 provide a review of current statistical analysis methodologies. A typical study design consists of 2 stages: first, an association analysis is carried out on each SNP, with the observed trait/phenotype status. Then the SNPs deemed most highly associated with the trait, and often additional surrounding SNPs are analyzed in independent samples in order to determine whether or not the signals are replicated.

The SNP data are analyzed in the case–control setting with one of a range of statistical tests, for example, Cochran–Armitage trend test, Cochran–Mantel–Haenszel test, logistic regression, and Bayes factors. Each of these tests assume certain genetic models and can account for the presence of population stratification and any additional information that may be available for the individuals in the study. The results from these analyses are usually in the form of a ranked list of *P*-values from which a subset of the top ranked SNPs is then taken forward for further study in an independent sample.

For each individual, as well as their genotype at each SNP, additional information is often recorded, for example, age, sex, and medical history. Also, as GWASs are often carried out in large consortia, with sample collections taking place at various sites, the site, and ethnicity information (sometimes self-reported or inferred) are also available. These are just some examples of extra data that may have been recorded, and we will henceforth refer to this type of additional information as additional phenotypic information (additional to the disease or trait status). This additional phenotypic information should, where possible, be incorporated into the testing for SNP association.

[0]To whom correspondence should be addressed.

In these analyses, all SNPs are treated in the same manner; in particular, they are all modeled as being equally likely to have the same impact on the phenotype. The exception being the Bayes factor approach which, among other advantages, allows for the inclusion of prior association information for each SNP. Although these analyses do sometimes incorporate additional phenotypic information such as that described above, rarely is other additional biological information included.

This biological information may be in the form of functional information, for example, biologically relevant location information for the SNPs, prior linkage findings, or prior association findings. This type of additional information that relates to the SNPs as opposed to the individuals in the study will be referred to as additional biological information. Approaches have been proposed to include this additional biological information in the form of multilevel or hierarchical models (Duditetal02,rma2003,LW2004,rieger2004).

Hierarchical or multilevel modeling, as the names suggest, consist of 2 or more levels that are organized in a hierarchical framework. The levels specify relationships between the variables and parameters that are of interest; these relationships may be in the form of regression equations, for example. The estimation of parameters in these models can be carried out in a fully Bayesian manner, where the distinction between data and prior information is strictly adhered to, with the prior distribution not depending on the data. A semi-Bayes approach can also be used in which prior information is subjectively fixed by the practitioner or an empirical Bayes approach in which the data themselves are used to specify priors. This latter approach can be thought of as lying somewhere between the fully Bayes and semi-Bayes approaches. Many authors in various fields have advocated the use of hierarchical models, and we suggest tomlins2005, TTC01 and Barlow$_{B}artholomew : 1972both for introductions to hierarchical models and for more formal and detailed accounts.$

Hierarchical modeling approaches offer many advantages over conventional methods. The ability to, easily and in a coherent framework, include relevant additional information—information that will help to inform on the disease/trait of interest—is one of the main advantages of this methodology. Hierarchical models offer better and more stable estimates of parameters than conventional methods. In general, estimates that are extreme and/or unstable become more reasonable in the hierarchical model setting and those parameter estimates that are more moderate remain so in the hierarchical approach (Gichangi$_{V}ach : 2006)$.

As previously mentioned, the *P*-value is the predominant measure used in ranking GWASs. Other authors have advocated using effect sizes rather than *P*-values in various settings, including in the detection of influential genetic markers (rieger2004,tomlins2005,TTC01). In the present context, the effect size can be thought of as indicating the "quantitative importance" (Gichangi$_{V}ach : 2006) of the association. On the other hand, the P-value does contain information on the magnitude of the effect size but this is confounded by the fact that it also contains inform values may have very different effect sizes as a result of having very different minor allele frequencies (MAFs). Thus, it is c values for ranking associations may not always be ideal. As well as being useful for the prediction of risk, the empirical Bayes h positive reduction.$

In the present context, the hierarchical modeling approach permits the relaxation of the assumption that all SNPs act similarly and allows for the SNP effects to vary, depending on the additional genetic information. Incorporating this additional information should help to control for false-positive associations that may be observed using conventional tests. Here, we present a 2-level hierarchical model and, depending on how the model is viewed, the interpretation of the role of the additional biological information may change. For example, in the frequentist setting, the multilevel model can be viewed as a mixed model with both fixed and random effects. If on the other hand, the model is viewed from a Bayesian perspective, the additional genetic information may be entering the model in the form of a prior distribution (see Section 1 of the supplementary material [SM] available at *Biostatistics* online).

Earlier hierarchical modeling approaches allowing for the incorporation of additional biological information for smaller scale association studies include Barlow$_{B}artholomew : 1972, who used a logistic regression model for the first level, in a$

Wahba:1990 provide a hierarchical regression modeling approach, the first level of which is based on the positive root of a 1-degree of freedom (df) χ^2 statistic, x_m, which has a noncentral χ distribution with 1 df and noncentrality parameter λ_m. The λ_m, which are of main interest, indicate the strength of association and in turn are modeled using a mixture model, allowing the incorporation of any prior knowledge of association. The prior depends on additional covariate information such as, for example, whether the marker is located in an exon, or a splice junction site, its predicted effect on protein conformation, or prior linkage, or association evidence. The authors compare various empirical and fully Bayesian schemes for this modeling framework using simulated data, where performance is assessed in terms of power.

Brown:1989 also propose a 2-level hierarchical modeling approach and explore both empirical Bayes and semi-Bayes approaches. For their empirical Bayes approach, they develop a structured weighting function that reflects the residual variation in the first-level estimates that remains after the second-level covariates have been taken into account. This weighting function is designed to give more support to particular markers based on one or more different types of additional biological information. For example, those markers with higher previous linkage or

association scores may be weighted more heavily. They apply their methods to both GWA SNP data and gene expression data.

$\mathrm{SAS}_{Institute}: 1999 presents both empirical Bayes and semi-Bayes adjustments that shrink the conventional effecti$
$level estimates arising from the Cochran--Armitage test for trend. The author also explores a semi-Bayes approach.$

Most of the analyses cited above consider the application of the empirical Bayes hierarchical model (EB-HM) to experimental data sets only and, even where they do include the use of additional covariate information, do not present any exploration of the sensitivity of the approach to this information. The exception is $\mathrm{Fimmers}_{Seuchter}:$
$1989 who in contrast do not present any application to experimental data but restrict their attention to simulation studies. In this$

We present a 2-level EB-HM for the analysis of individual marker GWA data, focusing on the case–control study design. This modeling framework consists of a first-level logistic regression model for each individual SNP with a second-level regression, incorporating additional biologically relevant information. An empirical Bayes approach, using techniques proposed by Barlow:1972, to iteratively estimate prior parameters in the model using the available data, is proposed. A key advantage of this empirical Bayes approach is that it does not rely on the subjective input of the practitioner in setting prior information but instead uses the available data to obtain parameter estimates. Previous examinations of the empirical Bayes approach have highlighted the conservative nature of the methodology, in particular when compared to the subjective semi-Bayes approaches ($\mathrm{Castelloe}_{Zimmerman}:$
$2002, Schwender: 2007). We acknowledge that our objective approach presented here is conservative and, in our implementat$
$-- as evidenced in both our simulation and experimental data applications.$

We explore a comprehensive simulation study that shows the effectiveness of our approach when truly relevant additional biological information is included. We also demonstrate, through simulations, the effect of applying this methodology when noisy, incomplete information is included. Although we do not incorporate linkage disequilibrium (LD) in our simulation studies, we do not feel this is a major drawback. In identifying variants that are associated with the phenotype of interest, the usual single-level approach can result in a set of markers being identified that are correlated with each other to varying degrees. Each of these variants is not usually treated as a separate association signal but rather the region is identified. If LD is of concern, the approach of LD pruning could first be employed—resulting in a set of independent, uncorrelated markers. Also, markers that are in LD with those in functional regions can be identified and such information incorporated in the additional covariate information, as we demonstrate in our experimental data applications. We also assess our proposed model extensively in terms of power and false-positive rates under various scenarios. We conclude by applying the methodology to 4 of the Wellcome Trust Case Control Consortium data sets: T2D, coronary artery disease (CAD), Crohn's disease (CD), and rheumatoid arthritis (RA) (allison2006).

2 Methods

In our development of the methodology for the hierarchical model, we will concentrate on the case–control study design. Here, we present a 2-level EB-HM. The first level comprises a logistic regression model, and the second level comprises a Gaussian regression incorporating additional biologically relevant information.

2.1 First level—logistic regression

The first level, in the 2-level hierarchical model for case–control type data, consists of the analysis of the individual SNPs; independently testing for associations between the individual SNP markers and case–control status. Given N individuals, a proportion of which will be cases and the remaining individuals controls, proportions that need not necessarily be equal, for each of the M SNPs we propose using a logistic regression model to test for association. The logistic regression model is chosen as it is a flexible model choice, allowing for the incorporation of various genetic models (e.g., additive, recessive, and dominant) which may be thought to underlie the disease or trait under consideration. The logistic regression approach also allows for the inclusion of additional phenotypic information through the inclusion of additional covariates. For the phenotype data (case–control status), for $i = 1, \ldots, N$ individuals, let $y_i = 1$, if individual i is a case, and let $y_i = 0$, if individual i is a control. For the genotype data, for $m = 1, \ldots, M$, we let X_{mi} denote the genotype for individual i at marker m. The genetic model may be incorporated here through the choice of coding for the number of minor alleles (from here on referred to as the allele) at the particular marker locus. For example, for an additive model, X_{mi}, takes the values 0, 1, or 2, indicating 0, 1, or 2 copies of the allele. For a dominant model, X_{mi} can take the values 0 or 1, 0 indicating 0 copies of the allele and 1 indicating one or 2 copies of the allele. Similarly, other genetic models may be incorporated.

The logistic regression model is given by the following equation:

$$\log\left(\frac{p_{mi}}{1-p_{mi}}\right) = \text{logit}(p_{mi}) = \alpha_m + \beta_m X_{mi}, \tag{1}$$

where p_{mi} is the probability of individual i being a case given that they have genotype X_{mi} at marker m. α_m, the baseline risk of disease, is the log odds for an individual having the disease given that they have the homozygous genotype for the major allele (i.e., they have 0 copies of the minor allele). The β_m are an estimate of the increase in odds of being a case for each additional allele (assuming additive model) on the log-odds scale and are thus an estimate of the effect size. Where available, additional phenotypic information, such as that previously mentioned, may be incorporated through the inclusion of additional covariates and their corresponding coefficients in (1), but for simplicity, we will not include these here. The statistical significance of the association can be tested using the Wald statistic, given by $\left(\hat{\beta}_m/\text{SE}(\hat{\beta}_m)\right)^2 \sim \chi^2$, where $\hat{\beta}_m$ is the maximum likelihood estimator (MLE) of β_m and $\text{SE}\left(\hat{\beta}_m\right)$ is its associated standard error. In the usual approach to analyzing GWAS data, the resulting *P*-values would be ranked in ascending order, and a *P*-value threshold (such as genome-wide significance; Duditetal02) used in order to determine which SNPs to investigate further. A logistic regression model has been used in rma2003, LW2004, and rieger2004, for example, to detect SNP associations in GWASs for various diseases. Our aim is to include additional biologically relevant information in our analysis through the incorporation of a second-level regression equation in our model.

2.2 Second level—regression

The second level of the model aims to improve the estimates of the β_m through the inclusion of additional biological information. This is facilitated through the inclusion of a second regression equation given by

$$\beta = Z\gamma + \tau^2 T, \quad \beta = (\beta_1, \ldots, \beta_M), \quad \beta_m \sim \text{Normal}\left(Z_m\gamma, \tau^2 t_{mm}\right), \tag{2}$$

where Z is an $M \times K$ matrix containing K columns of additional biological information. Thus, for each β_m (and thus each marker m), there is information on each of the additional biologically relevant covariates. There is no restriction on the values the entries in the matrix Z can take. For example, negative entries could be used to indicate that a particular covariate was expected to have a negative or opposite effect (protective) in contributing to susceptibility to disease. For the type of information included in Z, it will often be the case that many of the entries will be binary, indicating membership of a particular functional class, for example, SNPs in introns.

The vector γ is a $K \times 1$ vector of second-level regression coefficients. $\tau^2 T$ is a variance–covariance matrix representing the residual variation in the β_m's that remains after the second-level covariates in the Z matrix have been taken into consideration. Thus, $\tau^2 T$ reflects any unmodeled and unaccounted-for variability which could be due to, for example, interaction effects between the second-level covariates or covariates not included. Correlation structure between the SNPs would be modeled in the off-diagonal entries of the T matrix. If no correlation structure is to be modeled, which can reduce the computational burden, then T can be defined as the identity matrix. tomlins2005 follow a similar rationale in defining T.

In Section 1 of the SM available at *Biostatistics* online, we explore the mixed model and Bayesian interpretations of combining (1) and (2) which can help clarify the roles played by the various components of the model.

2.3 Combining first- and second-level effect estimates

We combine the first-level estimates and the second-level means for the β_m using the hierarchical estimator:

$$\hat{\beta}_{m;\text{EB}} = B_m Z_m \gamma + (1 - B_m)\,\hat{\beta}_m. \tag{3}$$

Equation (3) is sometimes referred to as a shrinkage estimator (TTC01) because the first level usual estimators, β_m, are shrunk toward, or averaged with, the second-level means $Z_m\gamma$. The shrinkage factor B_m indicates how much the second-level mean $Z_m\gamma$ contributes to the estimate β_m. Gichangi$_V ach$: $2006, inhisanalysis, setsZ_m\gamma$ to be zero for all SNPs, as it is felt that this could be a reasonable choice in GWAS applications. If the first-level estimators are highly variable, with respect to the total variability, they will be shrunk toward the more stable estimate, $Z_m\gamma$, thus reducing the variability. On the other hand, if the logistic regression estimates have small variability, then these estimates will not be shrunk as much toward the second-level means. The important question is how to determine the shrinkage factors $\{B_m\}$. There are various methods of choosing/estimating these, for example, empirical Bayes, semi-Bayes, and fully Bayesian approaches. Here, we will only concentrate on using an empirical Bayes estimator. Just as the means for the second-level model are estimated from the data, we will also estimate the prior means, and consequently, the shrinkage factors, using the data. This is in contrast to the semi-Bayes approach that would rely on the subjective judgement of the data analyst in setting the prior means.

We will use the empirical Bayes iterative approach described in Barlow$_B artholomew$: $1972 toestimatetheshrinkagefactors$.

2.4 Simulation strategy

To evaluate and explore the performance of the approach proposed here, we simulated case–control genotype data for a number of markers. The multilevel format of the model allows for a clear simulation framework, the natural starting point for the simulation being the second-level regression equation. Full details of the simulation strategy are given in Section 3 of the SM available at *Biostatistics* online.

3 Results

3.1 Simulation study application

To examine the efficiency of the proposed method for the better identification of the associated markers, we simulated case–control genotype data for 200 markers for each of 500 cases and 500 controls. We randomly simulated the MAFs on the interval $(0.01, 0.5)$ and chose a dominant genetic disease model with a disease prevalence of 0.05. Of the 200 markers, 3 markers were randomly chosen to have higher odds ratios than all other markers (markers 44, 123, and 184). The matrix Z contained 5 columns of additional covariate information, simulated from a Binomial distribution. The first column of the Z matrix was chosen as the most informative, and for the randomly chosen markers, the corresponding entries in this column of Z were increased. The aim was to detect these markers as associated markers, while also minimizing the number of false positives. The results of this simulation study can be seen in Figure 1. The residual variation that remains in the first-level coefficients, after the second-level additional covariate information is included, is modeled by τ^2 and was set at 0.01 in the simulation study. This is well estimated with the empirical Bayes approach as $\hat{\tau}_2 = 0.00723$. As can be seen in Figure 1, the EB-HM performs well in comparison to the MLEs of the logistic regression model, both in the detection of the markers with higher odds ratios and in the decrease of false-positive findings through the reduction of the standard errors.

We also compared the performance of the modeling framework with respect to estimating the true odds ratios over a number of random simulations. As noted by Wahba:1990, the effect estimates can be used for risk prediction and they give an indication of the quantitative importance of putative associations. To do this, we use the mean square error MSE $= M^{-1} \sum_m \left(\text{est(OR)} - \text{true(OR)}\right)^2$, where the est(OR) is either the odds ratio as estimated by the MLE from the logistic regression or the empirical Bayes estimate of the odds ratio, and the true(OR) is the simulated odds ratio. The MSE captures the degree to which the estimator differs from the true odds ratio, which can either be due to the stochastic nature of the process or the poor performance of the estimator. We examine the MSE across 100 random simulations for both the MLE and for the empirical Bayes hierarchical modeling estimators and compare them in Figure 1(a) of the SM available at *Biostatistics* online. We also plot the proportion of MAFs that are $< 5\%$ in each simulation. This is done to explore the possibility that simulations with many markers with low MAFs, which could lead to decreased power to detect associations, may be influential in the behavior of the estimators. There does not appear to be any evidence for this.

The simulation studies so far have used the same Z matrix, containing the same additional covariate information, in both the data simulation and the analysis phases. For experimental data, it will almost always be the case that such exact information will not be available. To examine the situation where additional noisy and incomplete information is provided in the analysis phase, we explored the scenario where a different Z matrix, $\hat{Z}$, is used in the analysis stage to that used in the simulation phase. In the simulation phase, Z is used, but in the analysis phase, we use $\hat{Z} = Z_{k-1} + e$, where $e \sim \text{Normal}(0, 1)$ and Z_{k-1} is the Z matrix with one column of extra covariate information omitted. All other parameters are similar to those in the simulation studies described above. The results of this simulation study can be seen in Figure 2. τ^2 is set again at 0.01, and $\hat{\tau}^2$ is estimated as 0.00584. The EB-HM still performs better than the single-level MLE estimates from the logistic regression. The aim would be that the empirical Bayes model should not perform less well than the single-level model, that is, that the additional noisy and incomplete information should not lead to dramatically different conclusions regarding which markers are believed to be associated. For this simulation scenario, we also simulated 100 random simulations and compared the MSE for both the empirical Bayes estimators and the MLEs. As can be seen in Figure 1(b) of the SM available at *Biostatistics* online, the empirical Bayes model still outperforms the single-level logistic regression model when considering the estimation of the odds ratios.

Finally, we show that the model does offer substantial improvements over the usual single-level model by examining the sensitivity and specificity of the model for simulated data. We use receiver operating characteristic (ROC) curves to assess these. In order to access the sensitivity (power) and specificity (1 false-positive rate), we generated 1000 random simulations, each having 2 random markers (out of 200) with higher odds ratios and used the same Z covariate matrix in the analysis and generation phases. We also carried out this same scenario for 15 random markers with higher odds ratios. As can be seen from the ROC curves in Figure 3(a) and (b), EB-HM performs substantially better than the MLE when relevant informative information is provided (Figure 3(a)

area under curve [AUC] MLE: 0.896, EB-HM: 0.998; Figure 3(b) AUC MLE: 0.896, EB-HM: 0.999). To assess the performance when noninformative information is provided in the Z covariate matrix, as may be the case in experimental analyses, we also carried out the same simulation strategy as just described but used positive random noise (abs[Normal$(0, 1)$]) for a reduced number of covariates (5 columns in Z in data generation phase, and 4 columns in analysis phase). Examining Figure 3(c) and (d), it is clear that the model is robust in the face of wrong, uninformative information and performs just as well as the MLE method, (Figure 3(c) AUC MLE: 0.896, EB-HM: 0.889; Figure 3(d) AUC MLE: 0.897, EB-HM: 0.903).

3.2 Application to WTCCC 2007 data sets

We present applications of our methodology to CAD, CD, RA, and T2D data sets from the [**?**]. Several of the genetic associations presented in Brown:1989 have been replicated. All data sets were quality controlled to include SNPs with MAF > 0.01 and a Hardy–Weinberg equilibrium threshold of $P < 10^{-4}$ in both cases and controls. This resulted in 2938 controls and CAD: 1926, CD: 1748, RA: 1860, and T2D: 1924 cases. We examined autosomal SNPs (CAD: 380442, CD: 380664, RA: 380481, and T2D: 380463) that had additional covariate information available.

The covariate information that constitutes the Z matrix for each of these data sets consisted of 20 categories of additional functional information for each SNP. Specifically: 11 binary categories (Downstream, Essential Splice Site, Intergenic, Intronic, Non-Synonomous Coding, Regulatory Region, Splice Site, Stop, Synonomous Coding, Upstream, and Untranslated Region (UTR)) and 9 linkage disequilibrium sum (LDSUM) categories (LDSUM Downstream, LDSUM Frameshift Coding, LDSUM Intergenic, LDSUM Intronic, LDSUM Non-Synonomous Coding, LDSUMStop, LDSUM Synonomous Coding, LDSUM Upstream, and LDSUM UTR). The scoring used for the binary covariate categories is as follows. For example, if an SNP scores a 1 for the intronic region, this indicates that the SNP is contained in an intron, similarly for the other binary categories. For a particular marker, the LDSUM categories denote the number of SNPs that are proxy SNPs (in LD) belonging to the various functional categories. For example, if an SNP has 4 proxies; 2 coding, 1 UTR, and 1 intergenic, then this SNP will score 2 on LDSUM Frameshift Coding, 1 on LDSUM UTR, and 1 on LDSUM Intergenic. The tool SNAP (Fimmers$_Seuchter$: $1989)wasusedtoidentifyproxySNPswithina500kbwindowusingHapMapCEUgenotypes.ThetoolGeneCruiser(Barlow$: $analysiscarriedoutbyCastelloe_Zimmerman$: $2002.Thismeta-analysisoriginallycontainedaUKsample, whichmayhave$ $analysisandthescoresrecalculated.Thelinkagescoresarefor115binsacrossthegenome(excludingthesexchromosome).Th$ $levellogisticregressionoddsratiosandstandarderrorswereobtainedusingPlink(Schwender$: $2007).Wemaketheassumpti$ that remains in the first-level coefficients after the second-level additional covariate information is included, is the same across all markers.

3.2.1 Coronary artery disease.

For CAD, the allison2006 found a notable new finding on chromosome 9p21.3 with the strongest signal seen at rs1333049. In both our first-level logistic regression analysis and in our EB-HM analysis, this marker is also our top signal (MLE OR $= 1.37$, *P*-value $= 2.3 \times 10^{-14}$, EB-HM OR $= 1.03$, *P*-value $= 0.016$). This signal has been replicated in a German sample (Duditetal02), showing a slightly reduced odds ratio $= 1.33$ (95% confidence interval [CI] 1.18–1.51). Two other loci on chromosome 6q25.1 (rs6922269) and on chromosome 2q36.3 (rs2943634) also showed moderate association with CAD in the rma2003 analysis and were replicated (LW2004). These markers also appear in our top list when ranking is carried out using the EB-HM approach, see Table **??** for further details. For the CAD data set, $\hat{\tau}^2 = 0.000199$.

3.2.2 Crohn's disease.

A number of loci that were previously identified as being associated with CD were also found to be associated in the tomlins2005 data and these also rank highly here based on our EB-HM approach (rs17221417, rs11805303, rs10210302, rs10761659, and rs17234657). A set of novel loci (rs1000113, rs9858542, rs10883365, and rs2542151) that have been replicated (TTC01) were also identified and these too rank highly in our EB-HM analysis. The top ranking marker based on MLE ranking is rs2076756 on chromosome 16. This marker has been replicated (Gichangi$_Vach$: $2006)andinourEB-HMrankingappearsinthetop150markers.Unliketheother3diseasedatasets, wehaved$ $HMresults, thisisnotthecasewiththeCDdataset.Here, thetop120rankedmarkersbasedonEB-HMrankingallcomefrom$ $-associatedloci(Barlow_Bartholomew$: $1972).FortheCDdataset,^2 = 0.00036$.

3.2.3 Rheumatoid arthritis.

Previous associations between RA and the human leukocyte antigen region and the PTPN22 gene on chromosome 1p13 were also found in the WTCCC (2007) analysis. The most associated marker for PTPN22 was rs6679677

(MLE OR $= 1.95$, *P*-value $= 2.86 \times 10^{-25}$) and is also highly ranked in the EB-HM analysis (EB-HM OR $= 1.04$, *P*-value $= 0.01$). The top ranking marker in the first-level analysis (rs6457617, MLE OR $= 2.26$) on chromosome 6 is also the top ranking marker in the EB-HM analysis (EB-HM OR $= 1.1$). See Table **??** for further details. For the RA data set, $\hat{\tau}^2 = 0.00027$.

3.2.4 Type 2 diabetes.

Previously identified loci on chromosomes 3, 10, and 11 were replicated in the Wahba:1990 study (rs4506565, rs1801282, and rs5215). Marker rs4506565 is ranked first by both the MLE and the EB-HM approaches, and rs1801282 and rs5215 appear further down in both rankings. Two other signals were also identified on chromosome 16q (rs8050136, rs9939609, and rs7193144) and on chromosome 6p22 (rs9465871) and all associations are highly ranked with both the MLE and the EB-HM approaches as can be seen in Table **??**. For the RA data set, $\hat{\tau}^2 = 0.000234$.

In Figure 4, we have plotted the MLE odds ratios from the first-level logistic regression model (first level) and the empirical Bayes odds ratios. The reduction in effect sizes in EB-HM markers compared to the MLE odds ratios can clearly be seen. But previously identified signals appear highly ranked when considering EB-HM ranking (examples in Table **??**, black points in Figure 4 plots). Figure 2 of the SM available at *Biostatistics* online presents details of the top 200 markers, using EB-HM ranking, as they appear along the genome (not-scaled) for each of the 4 diseases.

4 Discussion

In this paper we explore an empirical-Bayes two level hierarchical model that aims to better detect associated markers and provide more robust effect estimates in GWASs than the usual single level analysis typically carried out. This is done through a hierarchical modeling framework that allows the inclusion of additional biological information that is relevant, such as functional information or prior linkage or association information. This additional relevant biological information is included in a structured framework through a second-level regression equation. An existing iterative empirical Bayes method is used to assign prior means in the model and thus eliminates any subjective input that might be required by the practitioner, such as in semi-Bayes approaches.

Our aim was to explore how well this modeling framework worked with regard to detecting associations and obtaining robust effect estimates. In order to do this, we first examined the method in an extensive case–control simulation study that demonstrates how effective the method is when truly relevant additional biological information is incorporated—improving the *P*-value significance and giving better estimates of the odds ratios and reducing the standard errors, as can be seen in Figure 1. We also assessed the approach in terms of power and false-positive rates, under various scenarios, showing that when relevant informative covariate information is incorporated, the EB-HM approach performs better than the usual single-level model, see Figure 3(a) and (b). We also demonstrate, through simulation, that the EB-HM still performs well even when relevant information is omitted and noisy information is incorporated, although there is considerable reduction in the size of the odds ratios and in the level of significance of the *P*-values (see Figure 2). When uninformative noise is incorporated, the EB-HM approach performs as well as the single-level model in terms of power and false-positive rates, see Figure 3(c) and (d). This is an important demonstration as in real experimental applications it will rarely, if ever, be known how relevant the additional covariate information is. Thus, we show that our methodology is not adversely affected by the inclusion of unreliable additional data thereby ensuring robustness for practitioners when considering incorporating additional biological information.

The simulation studies do attempt to model as closely as possible the situation in experimental GWA data, but we acknowledge that there is no modeling of LD, as the genotypes for each individual are simulated independently. Nevertheless, as discussed in Section 1, we feel that this should not impact on the validity of the evaluation of the proposed modeling framework. The simulation studies also contain many fewer markers than would be contained in a GWAS, but this is only for demonstration purposes and could easily be extended to larger numbers of markers. Also, here we have concerned ourselves with the case–control study design, but it would be just as easy to consider the hierarchical model framework for family trio GWA data sets. For this study design at the first level, we would perform a logistic regression on the usual transmission disequilibrium data.

Previous applications of this type of modeling framework have not presented results for full GWA data sets that include additional covariate information that is believed to be informative. Thus, we have applied this methodology to 4 of the Brown:1989 data sets and included additional covariate information in the form of both functional information and prior linkage information (T2D only) that we believed would be informative. Across all 4 data sets, there is a reduction in the odds ratios and also the *P*-values have been increased considerably due to the reduction in the EB-HM odds ratios and their corresponding standard errors. Importantly, however is that in the

ranking of the association results based on the EB-HM approach, all previous identified and replicated signals are still highly ranked, agreeing with the MLE rankings, see Table **??**. The EB-HM analysis of the CD data set does result in the top ranking of a region on chromosome 17q21 that had not been identified in the $SAS_{Institute}$: $1999study, whilestillmaintainingallpreviouslyassociatedandreplicatedassociationsasalsohighlyranked.Thetoprankedn$ $values, correspondtothehighestempiricalBayesoddsratios, whichisnotthecaseforthe MLE P-valuerankedoddsratiosasc$ $levelestimatesareoftenoverinflated.Ascanbeseeninthesimulationstudies, theempiricalBayesapproachdoesnotsufferf$

We compared the T2D results obtained here with those obtained by Barlow:1972, who analyzed an almost identical WTCCC T2D data set (same number of cases and controls but slight difference in number of SNPs) using an empirical Bayes methodology, but who included no additional biological information, instead setting the prior mean effect for all SNPs to be zero. Our results did not differ considerably from Strömberg's when we included what we believed to be relevant covariate information. In particular, the odds ratios and standard errors of our estimates do not appear to be very different (comparison of our results with figure presented in $Castelloe_{Zimmerman}: 2002).Alsoallison2006obtainedа^2 = 0.00022$, which represents the residual variation that remains in the first-level coefficients, after the second-level covariates have been taken into account. This is almost identical to our estimate of $\hat{\tau}^2 = 0.000234$.

As pointed out by Duditetal02 and rma2003, among others, the empirical Bayes approach can sometimes be too conservative, resulting from too much shrinkage. Here, this conservatism manifests itself in the effect sizes, that is, the odds ratios of the EB-HM perhaps being underestimated and being closer to 1 than is actually the case. There is evidence for this in the simulation studies presented here and is also likely in the LW2004 applications, although the true effect sizes are often not known. For example, for the T2D data set, marker rs8050136 has an MLE OR $= 1.27$ and an EB-HM estimated OR $= 1.03$. In replication studies (rieger2004), this OR does vary substantially from 1.27, ranging from 1.03 (95% CI: 0.91–1.17) to 1.22 (95% CI: 1.12–1.32), and a combined sample OR $= 1.17$ (95% CI: 1.12–1.22). Thus, it would appear that even in replication studies a range of effect sizes can be observed, and it can be difficult with experimental data applications to judge how conservative the effect estimates are. For the simulation studies, the conservative behavior can be seen in particular in Figures 1(b) and 2(b), where the odds ratios are typically to be found in narrower ranges than the true simulated odds ratios. This excessive shrinkage and conservative nature of the empirical Bayes approach is often a reason why semi-Bayes approaches are preferred. Both tomlins2005 and TTC01 also consider semi-Bayes approaches, where the amount of shrinkage can be hand tuned. We feel this can be too subjective, as there is no available information to guide the choice of shrinkage in scenarios outside of simulation studies. The empirical Bayes approach offers the advantage of the practitioner not having to set parameter values, which may be more attractive in situations where limited knowledge is available.

The empirical Bayes approach to analyzing the $Gichangi_{Vach}: 2006datasetshasnotresultedinP-valuesofgenome-$ $widesignificance, butthisshouldnotleadtotheconclusionthatthemethodisinapplicable.Anumberofimportantpointsneedto$ $asnotedabove, theempiricalBayesapproachisknowntobeconservativeinitsestimatesofassociationandisdesignedmoretos$ $HMapproachisconsidered, allsignalspreviouslyidentified(Barlow_{Bartholomew}: 1972)andreplicatedappearhighlyrank$ $levelmodelandthismaynotbeadequatefortheincorporationofthisinformation.Itmayalsobethecasethatsufficientinform$ $levelcovariateinformationaddslittleindeterminingassociation.Becauseofourextensiveexamination, usingsimulations$ $positiverates, webelievethattheinclusionofthecovariateinformationisnothavinganegativeimpactontherankingofthem$

5 Software

Software in the form of R code, together with a sample input data set and complete documentation is available on request from the corresponding author (eaheron@tcd.ie).

6 Supplementary Material

Supplementary material is available online at `http://biostatistics.oxfordjournals.org`.

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A subset of the strongest associated markers for each of the 4 rieger2004 data sets (CAD, CD, RA, and T2D). MLE-OR(SE) refers to the logistic regression maximum likelihood odds ratio estimate and the standard error of the $\log(\text{MLE OR})$ as calculated for the analysis carried out here. MLE rank refers to the ranking of the marker based on the MLE *P*-value. EB-HM-OR(SE) refers to the EB-HM odds ratio and the standard error of the $\log(\text{EB-HM OR})$. The EB-HM Rank refers to the rank of the marker based on the EB-HM *P*-value and the effect size (ES) Rank refers to the ranking based on the EB-HM odds ratios

Dis	Chr	Marker	MLE-OR (SE)	MLE *P*-value	MLE rank	EB-HM-OR (SE)	EB-HM *P*-value	EB-HM rank	ES rank CAD
2	rs2943634	1.22(0.04)	1.23×10^{-5}	32	1.02(0.01)	0.19	28	37	
CAD	6	rs6922269	1.23(0.05)	6.58×10^{-6}	22	1.02(0.01)	0.2	37	44
CAD	9	rs1333049	1.37(0.04)	2.3×10^{-14}	1	1.03(0.01)	0.016	1	1
CD	1	rs11805303	1.37(0.04)	8.1×10^{-13}	7	1.05(0.02)	0.003	120	120
CD	2	rs10210302	1.39(0.04)	9.1×10^{-14}	2	1.05(0.02)	0.004	123	123
CD	3	rs9858542	1.26(0.05)	8.16×10^{-7}	71	1.03(0.02)	0.1	313	308
CD	5	rs1000113	1.52(0.08)	6.38×10^{-8}	48	1.03(0.02)	0.169	701	512
CD	5	rs17234657	1.54(0.06)	3.34×10^{-13}	5	1.04(0.02)	0.02	147	154
CD	10	rs10761659	1.24(0.04)	2.8×10^{-7}	61	1.03(0.02)	0.05	221	211
CD	10	rs10883365	1.28(0.04)	1.53×10^{-8}	30	1.04(0.02)	0.02	142	152
CD	16	rs17221417	1.36(0.04)	1.14×10^{-11}	16	1.05(0.02)	0.008	128	128
CD	16	rs2076756	1.44(0.05)	8.35×10^{-15}	1	1.05(0.02)	0.003	121	121
CD	17	rs916793	1.2(0.05)	0.0003	490	1.09(0.02)	1.68×10^{-6}	2	2
CD	17	rs17691328	1.21(0.05)	0.0003	448	1.09(0.02)	1.58×10^{-6}	1	1
CD	18	rs2542151	1.35(0.05)	5.1×10^{-8}	44	1.03(0.02)	0.06	235	230
RA	1	rs6679677	1.95(0.06)	2.86×10^{-25}	32	1.04(0.02)	0.01	52	52
RA	6	rs6457617	2.26(0.05)	2.17×10^{-72}	1	1.1(0.02)	1.38×10^{-9}	1	1
T2D	3	rs1801282	1.24(0.07)	0.0013	761	1.01(0.01)	0.38	2826	2228
T2D	6	rs9465871	1.29(0.05)	1.09×10^{-6}	19	1.02(0.01)	0.15	42	42
T2D	10	rs4506565	1.37(0.04)	7.1×10^{-13}	1	1.04(0.01)	0.02	1	1
T2D	11	rs5215	1.15(0.04)	0.00129	763	1.01(0.01)	0.347	1779	1787
T2D	16	rs7193144	1.27(0.04)	1.56×10^{-8}	11	1.03(0.01)	0.048	11	11
T2D	16	rs8050136	1.27(0.04)	2.16×10^{-8}	12	1.03(0.01)	0.05	12	12
T2D	16	rs9939609	1.26(0.04)	5.6×10^{-8}	15	1.03(0.01)	0.06	15	15

Figure 1: Results of a simulation study with 200 markers, for each of 500 cases and 500 controls. Markers 44, 123, and 184 are 3 random markers chosen to have higher odds ratios, and these are indicated in black, all other markers are in gray. The MAF for each marker was randomly simulated from Uniform(0.01, 0.5). The same Z matrix was used in both the simulation and analysis phase. (a) For both the MLEs and the empirical Bayes estimators (EB-HM), $-\log_{10}(P\text{-value})$ are shown. (b) The true odds ratios used in the simulation study together with MLE and empirical Bayes estimators of the odds ratios. (c) Confidence intervals for the logistic regression MLEs of the odds ratios. (d) Approximate confidence intervals for the empirical-Bayes estimated odds ratios.

Figure 2: Results of a simulation study with 200 markers, for each of 500 cases and 500 controls. Markers 12, 98, and 182 are 3 random markers chosen to have higher odds ratios and these are indicated in black, all other markers are in grey. The MAF for each marker was randomly simulated from Uniform$(0.01, 0.5)$. An alternative Z matrix was used in the analysis phase to that used in the simulation phase. (a) For both the MLEs and the empirical Bayes estimators (EB-HM), $-\log_{10}(P\text{-value})$ are shown. (b) The true odds ratios used in the simulation study together with MLEs and empirical Bayes estimators of the odds ratios. (c) Confidence intervals for the logistic regression MLEs of the odds ratios. (d) Approximate confidence intervals for the empirical-Bayes estimated odds ratios.

Figure 3: (a) ROC curves for both MLE estimates and EB-HM estimates based on 1000 random simulations of 500 cases and 500 controls, 200 markers with 2 causative markers. Same Z covariate matrix used in both generation and analysis phases. (b) Same as plot (a) except 15 causative markers are simulated. (c) Same as (a) except Z is random positive noise (abs[Normal$(0, 1)$]) in the analysis phase. (d) Same as plot (c) except 15 causative markers are simulated.

Figure 4: MLE of the odds ratios (first-level logistic regression model) and the empirical Bayes estimators (EB-HM) of the odds ratios for the autosomal markers (gray points) for (a) CAD, (b) CD, (c) RA, and (d) T2D. Points in black are the markers detailed in Table **??** for each of the disease data sets.